

Activity

21/7/26

1. Medical Image Enhancement

① which gray level transformation would you recommend

A. power Law (Gamma) transformation with $n < 1$

② why is it suitable

A. → It enhances the darker gray levels where soft tissues are present

→ It improves the visibility of low intensity regions without significantly affecting brighter bone regions

③ what effect will it have on the image

A. → Dark tissues become brighter and more visible

→ Bone details remain clear.

→ Overall image contrast improves, making diagnosis easier.

2. Night - Time Surveillance

① which transformation would improve the visibility

A. log transformation.

② why is it more suitable than negative transformation?

A → log transformation expands low-intensity pixel values (0-40), revealing details hidden in dark regions.

→ A negative transformation only inverts brightness and does not effectively enhance dark region details.

③ predict the appearance of enhanced image.

→ parking lot becomes brighter.

→ vehicles, people, and objects are easier to identify.

3. Fingerprint Recognition

① which gray-level transformation would you apply.

A. Contrast Stretching or power law transformation

② Explain why

→ It increases the contrast between fingerprint ridges and valleys.

→ It improves visibility of fingerprint ridges and valley by enhancing

③ what would happen if log transformation were applied instead

It would brighten the darker regions but may not improve the ridge valley contrast as effectively as gamma transformation.